

★ SHANNON ENTROPY IN HCI

- Shannon Entropy measures **uncertainty or unpredictability in information**.
- In HCI, it **quantifies how predictable user choices are in menus, buttons, or system responses**.
- Measured in **bits**.
- **High entropy** → **high uncertainty** → **harder decision**.
- **Low entropy** → **more predictable** → **easier interaction**.

★ MATHEMATICAL DEFINITION

$$H = - \sum p(x) \log_2 p(x)$$

- $p(x)$ = **probability of each possible event or user choice**
- $\log_2$ is used because information is **measured in bits**

★ INTUITION

- If **all options are equally likely** → **maximum entropy**
- If **one option is dominant** → **lower entropy**.
- If **one option has probability 1** → **entropy = 0**

More choices + equal probability → more uncertainty.

★ WHY SHANNON ENTROPY MATTERS IN HCI

- **Measures uncertainty in user decisions**.
- **Evaluates efficiency** of menu design.
- Used in **predictive interfaces**.
- Connected to Hick's Law (decision time increases with number of choices).
- **Lower entropy** → **faster interaction**
- **Good interface design** **reduces entropy by making choices predictable and clear**.

Key idea:

A well-designed menu with one frequently used option has lower entropy than a menu where all options are equally confusing.

★ NUMERICAL EXAMPLE – 4 MENU OPTIONS

A software interface has 4 options: A, B, C, D.

★ CASE 1 – EQUAL PROBABILITIES (HIGH UNCERTAINTY)

Option	$p(x)$	$\log_2 p(x)$	$p(x) \times \log_2 p(x)$
A	0.25	-2	-0.50
B	0.25	-2	-0.50
C	0.25	-2	-0.50
D	0.25	-2	-0.50

Step 1: Write formula

$$H = - \sum p(x) \log_2 p(x)$$

$$H = -[4 \times (0.25 \log_2 0.25)]$$

Step 2: Compute log

$$\log_2 0.25 = \log_2(1/4) = -2$$

Step 3: Multiply

$$0.25 \times (-2) = -0.50$$

$$4 \times (-0.50) = -2.0$$

Step 4: Apply negative sign

$$H = -(-2.0) = 2.0 \text{ bits}$$

Interpretation

- $H = 2$ bits
- This is the maximum entropy for 4 options.
- All options equally likely.
- User faces maximum uncertainty.
- Decision-making is slower.
- Interface is harder to use.

[What is the MAX ENTROPY????????]

- Entropy is maximum when all outcomes are equally likely.
- For n equally likely outcomes, maximum entropy is:

$$H_{max} = \log_2(n)$$

★ APPLYING THE RULE TO 4 OPTIONS

We have:

- Number of options = 4
- All probabilities = 0.25

Using formula:

$$H_{max} = \log_2(4)$$

$$\log_2(4) = 2$$

- ✓ So maximum possible entropy = 2 bits

That's why in Case 1, when all options were equal, we got 2 bits.

★ CASE 2 – UNEQUAL PROBABILITIES (LOWER UNCERTAINTY)

Option	p(x)	$\log_2 p(x)$	$p(x) \times \log_2 p(x)$
A	0.70	-0.515	-0.3605
B	0.10	-3.322	-0.3322
C	0.10	-3.322	-0.3322
D	0.10	-3.322	-0.3322

1 $H = -[0.70 \times \log_2(0.70) + 3 \times (0.10 \times \log_2(0.10))]$

2 $\log_2(0.70) \approx -0.515 \mid \log_2(0.10) \approx -3.322$

3 $H = -[(0.70 \times -0.515) + 3 \times (0.10 \times -3.322)]$

4 $H = -[-0.3605 + 3 \times (-0.3322)] = -[-0.3605 + (-0.9966)]$

5 $H = -(-1.3571) = \approx 1.36 \text{ bits}$

Interpretation

- $H \approx 1.36$ bits
- Lower than 2 bits.
- One dominant option (70%).
- Interface is more predictable.
- Faster interaction.
- Easier decision-making.

★ FINAL EXAM TAKEAWAY

- Maximum entropy occurs when all options are equally likely.
- Entropy decreases when one option becomes dominant.
- Lower entropy → lower uncertainty → faster decisions.
- Good interface design reduces entropy by guiding user expectations.